

INTERNAL TEAM REFERENCE — ALL PROJECTS

Network Consultancy

Web Accessibility

Widget Standard

A complete guide to the built-in accessibility widget — what it does, what's included, and why it's the default for every project we build.

DOCUMENT DATE

26 June 2026

WIDGET VERSION

v2.0

COMPLIANCE TARGET

WCAG 2.2 Level AA

What Is the Accessibility Widget?

Every website we build ships with a built-in, self-contained **Accessibility Widget** — a floating panel that users can open at any time to adjust how the website looks and behaves to suit their personal needs.

It lives in the bottom-right corner of every page as a small blue circular button. Clicking it opens a full-height side panel with grouped settings across four categories: *Profiles*, *Content*, *Colour*, and *Navigation*.

DEFAULT STANDARD — ALL PROJECTS

This widget is baked into every project we deliver from this point forward. It requires no third-party services, no subscriptions, no CDN dependencies, and no cookies. It is 100% self-hosted, pure HTML/CSS/JavaScript.



Current Status — Live & Active · v2.0 Deployed across the Network Consultancy website (netcon-ivory.vercel.app). Last updated 26 June 2026. All settings persist in the user's browser via localStorage.

1 Why We Include It By Default

- **Legal compliance:** The UK Equality Act 2010 and the Public Sector Bodies Accessibility Regulations require reasonable adjustments for users with disabilities. Including an accessibility panel demonstrates clear commitment.
- **User reach:** Approximately 1 in 5 people in the UK have some form of disability. An accessible site serves more visitors, reduces bounce rate, and improves conversions.
- **SEO & Core Web Vitals:** Semantic, accessible markup is rewarded by Google's ranking algorithms.
- **No extra cost:** Zero licensing fees. No script tags from third parties. No GDPR risk from external analytics.
- **Client confidence:** Clients can see a tangible, visible accessibility feature on their site. It builds trust.
- **Future-proof:** WCAG 2.2 Level AA is currently the accepted standard. The widget is already structured to extend to WCAG 3.0 as it matures.

2 How It Works — Quick Technical Summary

Two files power the entire widget. They are included on every page via a standard HTML link/script tag pair:

```
<!-- In <head> -->
<link rel="stylesheet" href="/css/accessibility.css">

<!-- Before </body> -->
<script src="/js/accessibility.js" defer></script>
```

The JavaScript injects the widget's HTML into the page at runtime — so there is nothing to copy into individual page files. The widget appears automatically on any page that loads both files.

User preferences are saved to **localStorage** under the key `nc_a11y_v2`. Settings persist across page navigation and browser sessions until the user resets them.

SECTION A

Accessibility Profiles

Profiles are **one-tap presets** that activate a combination of settings tailored to a specific accessibility need. They are designed for users who don't want to adjust individual settings manually. Profiles do not block individual settings — users can still fine-tune on top of a profile.

Epilepsy Safe Mode

Designed for users with photosensitive epilepsy. Removes all flashing, animated, and visually intense content.

Activates automatically: Pause Animations + Low Saturation

Seizure Safe Mode

For users prone to seizure triggers. Combines motion removal with a calmer dark visual theme.

Activates automatically: Pause Animations + Dark Contrast

ADHD Friendly

Aids focus and readability for users with ADHD or attention difficulties. Improves spacing, rhythm, and visual anchoring.

Activates automatically: Line Height + Letter Spacing + Highlight Links

Low Vision

For users with reduced visual acuity. Maximises legibility by increasing text size and contrast to its most readable state.

Activates automatically: High Contrast + Font Size at 125%

IMPORTANT — HOW PROFILES INTERACT WITH INDIVIDUAL SETTINGS

Profiles compute an "effective" state on top of whatever the user has individually set. For example: if a user already has High Saturation turned on and then activates Epilepsy Safe (which enables Low Saturation), the profile's setting takes priority while the profile is active. Deactivating the profile restores the user's prior individual state.

SECTION B

Content Adjustments

These settings let users control how text and links are presented. All of them apply to page content only — the accessibility widget itself is always excluded from these overrides to ensure it remains usable.

FEATURE	STATUS	WHAT IT DOES
Font Size	EXISTING	Scales the entire page up or down in steps from 75% to 175%. Shows current percentage. Adjust with + / – buttons.
Highlight Titles	NEW	Adds a dashed blue outline and subtle background tint to all headings (H1–H6). Helps users locate sections quickly.
Highlight Links	EXISTING	Highlights every hyperlink with a yellow background and black border. Particularly useful for users with colour vision deficiency.
Dyslexia Font	EXISTING	Switches all body text to Arial/Verdana with wider letter and word spacing. Research shows these changes reduce reading errors for dyslexic users.
Letter Spacing	NEW	Increases the space between characters across all text. Applies to paragraphs, list items, buttons, table cells, and labels.
Line Height	NEW	Increases vertical spacing between lines of text to 2.0× the default. Reduces visual crowding and aids reading flow.
Font Weight	NEW	Forces all body text to bold (weight 700). Useful for low-contrast displays or users with mild visual impairment who need stronger text definition.
Text Align	NEW	Cycles text alignment through Left → Centre → Right and back to default. Some users find left-aligned text easier to read; others prefer centred.

SECTION C

Colour Adjustments

These settings modify how colours are rendered. **Contrast modes** (Dark, Light, High) are mutually exclusive — enabling one disables the others. **Saturation modes** (High, Low, Monochrome) are also mutually exclusive.

FEATURE	STATUS	WHAT IT DOES	GROUP
Dark Contrast	NEW	Forces a dark navy/charcoal theme across the page. Good for users sensitive to bright white screens or working in low-light conditions.	Contrast
Light Contrast	NEW	Applies a brightness + contrast boost via CSS filter. Brightens the page for users who struggle with low-contrast designs.	Contrast
High Contrast	EXISTING	Extreme black-and-white mode. All backgrounds become black, all text becomes white, links become yellow. Maximum legibility for severely impaired vision.	Contrast
High Saturation	NEW	Boosts colour saturation to 250% via CSS filter. Helps users with mild colour vision deficiency distinguish between colours more easily.	Saturation
Low Saturation	NEW	Reduces colour saturation to 30%. Calmer, less visually stimulating. Also activated by the Epilepsy Safe profile.	Saturation
Monochrome	EXISTING	Converts the entire page to greyscale via CSS filter. Useful for users with full colour blindness or those testing colour-independence.	Saturation

SECTION D

Navigation Adjustments

These settings affect how users interact with and navigate the page — beyond just the visual appearance.

FEATURE	STATUS	WHAT IT DOES
Mute Sounds	NEW	Immediately mutes all <audio> and <video> elements on the page. Useful for hearing-sensitive users or screen-reader users who don't want competing audio.
Page Read	NEW	Activates hover-to-speak using the browser's built-in Web Speech API. When active, hovering over any paragraph, heading, link, or button reads its text aloud in a British English voice (en-GB). Works in all modern browsers.
Reading Guide	NEW	Displays a horizontal semi-transparent blue band that follows the user's mouse cursor vertically — like a physical ruler held under a line of text. Helps users with tracking difficulties (dyslexia, ADHD) keep their place.
Pause Animations	EXISTING	Instantly stops all CSS animations, transitions, and scroll behaviours site-wide. Also activated by the Epilepsy Safe and Seizure Safe profiles. Equivalent to the OS-level "Reduce Motion" preference.
Big Cursor	NEW	Replaces the browser's default cursor with a large, high-contrast black-and-white arrow cursor (40×40px). Helps users who have difficulty locating a small pointer on screen.

Compliance & Standards

A WCAG 2.2 Level AA

The widget is built to satisfy the following WCAG 2.2 success criteria directly:

- **1.4.3 Contrast (Minimum)** — High Contrast mode
- **1.4.4 Resize Text** — Font Size control
- **1.4.10 Reflow** — Zoom-based scaling preserves layout
- **1.4.12 Text Spacing** — Letter Spacing, Line Height
- **2.1.1 Keyboard** — Full keyboard navigation, Tab/Shift+Tab, Escape to close
- **2.4.1 Bypass Blocks** — Skip navigation link present on all pages
- **2.4.7 Focus Visible** — Custom focus ring on all interactive elements
- **2.3.3 Animation from Interactions** — Pause Animations
- **4.1.3 Status Messages** — Font size % shown in live region

B ARIA Roles

The widget uses proper ARIA markup throughout:

- Trigger: `aria-expanded`, `aria-haspopup="dialog"`
- Panel: `role="dialog"`, `aria-modal="true"`
- All buttons: `aria-pressed` (toggled on/off)
- Font % display: `aria-live="polite"`
- Icons: `aria-hidden="true"`
- Sections: `role="group"` with `aria-label`
- Focus trap: Tab/Shift+Tab contained within open panel

File Reference

FILE	PURPOSE	SIZE
<code>css/accessibility.css</code>	Widget UI styles (panel, buttons, layout) + all mode override classes applied to body	736 lines
<code>js/accessibility.js</code>	Widget logic: state management, DOM injection, all toggle handlers, TTS, reading guide, focus trap, localStorage persistence	635 lines

USING ON PAGES IN SUBDIRECTORIES

Pages inside `services/`, `solutions/`, `industries/`, `resources/`, `case-studies/` must use root-relative paths: `/css/accessibility.css` and `/js/accessibility.js` — or adjust with the correct `../` prefix relative to the file's depth.

Team Notes & Project Policy

- **Include on every project:** Both files must be included on every page of every site we build, regardless of project size or client type.
- **No modifications without version bump:** If you change the JS or CSS, update the `STORAGE_KEY` in `accessibility.js` (e.g. `nc_a11y_v3`) to avoid conflicts with saved user settings from the previous version.
- **Keep the widget isolated:** All CSS overrides use `:not(#a11y-widget):not(#a11y-widget *)` selectors to prevent accessibility modes from styling the widget itself and breaking usability.
- **No colour inversion in v2.0:** The old "Negative Contrast" (CSS `invert(1) hue-rotate(180deg)` filter) was removed and replaced with the separate Dark/Light/High Contrast modes. It can be re-added as an individual button in the Colour section if needed.
- **Page Read requires user gesture:** Browser autoplay policies mean the Web Speech API (`speechSynthesis`) will only work if the user has interacted with the page first (i.e. they clicked the widget button to enable it). This is expected browser behaviour, not a bug.
- **Big Cursor cross-browser note:** SVG data URI cursors are supported in all modern browsers (Chrome, Edge, Firefox, Safari). If a client's brief specifies IE11 support, a fallback cursor approach should be discussed.

VERSION HISTORY

Update Log

VERSION	DATE	SUMMARY OF CHANGES
v2.0	26 Jun 2026	Complete rebuild. Added 4 Accessibility Profiles, 5 new Content features (Highlight Titles, Letter Spacing, Line Height, Font Weight, Text Align), 3 new Colour modes (Dark/Light Contrast, High/Low Saturation), 4 new Navigation features (Mute Sounds, Page Read, Reading Guide, Big Cursor). Redesigned panel from popup to full-height side drawer.
v1.0	— 2026	Initial release. 8 settings: Font Size, High Contrast, Negative Contrast, Greyscale, Dyslexia Font, Highlight Links, Increase Spacing, Pause Animations. Compact 320px popup panel.